

Immunohistochemistry on paraffin sections

Little Group, GUDMAP Consortium

We use kits for IHC:

For mouse primary monoclonal and polyclonal antibodies on mouse tissues:

VECTOR Laboratories VECTOR M.O.M. Immunodetection Kit
BASIC Catalog No. BMK-2202

We have in the past used VECTASTAIN Elite ABC KITS specific to the secondary antibody needed, but we are now using our own blocking buffer and variously sourced secondary biotin-conjugated antibodies.

For the horseradish peroxidase we use:

VECTOR Laboratories VECTASTAIN Elite ABC Reagent PK-6100

For the colour reaction we use:

VECTOR Laboratories DAB Substrate Kit for Peroxidase SK-4100

FOR NEW WAX SECTIONS;

Deparaffinise and rehydrate sections

In coplin jars (10 slides) (50ml) or square glass jars (>10slides) (300ml)

2 X 3min	xylene in fume hood
2 X 2min	100% ethanol in fume hood
1 X 2min	95% ethanol
1 X 2min	80% ethanol
1 X 2min	70% ethanol
1 X 5min	PBS (or water if blocking with H ₂ O ₂)

You can keep your solutions of xylene and ethanol and re-use them up to 10 times.

Proteinase K Antigen Retrieval/Unmasking

KYLIE: I don't use this step for IHC on top of RNA ISH.

- Make a fresh solution of 25ul 20mg/ml Proteinase K
2.5ml 1M Tris-HCl, pH 8.0
0.5ml 0.5M EDTA, pH 8.0
up to 50ml with milliQ water
- Incubate slides in this solution at 37C for 5min in a coplin jar
- Wash 3 X 5min PBS

Quenching/Blocking endogenous peroxidase activity*

*Perform especially if there is a problem with background due to endogenous peroxidases, eg in adult kidney, the incubation times may be shortened by using higher concentrations of H₂O₂. If endogenous peroxidase activity does not present a problem, omit this step.

- Incubate 30min 0.3% H₂O₂ in tap water (1/100 of a 30% stock solution, kept at 4C) OR incubate 3min 3% H₂O₂ in tap water
- Rinse 3 X 5min PBS

(For frozen sections: incubate 5min 0.3% H₂O₂ in 0.3% serum in PBS)

FOR IHC ON TOP OF RNA ISH START HERE;

NOTE; It is best to do the IHC straight away as in some cases we have seen the stain from ISH reduce if left too long in PBS. But O/N in PBS 4°C should be ok.

Blocking

The preferred serum for blocking is prepared from the same species in which the biotinylated secondary antibody is made.

- Tip off excess PBS and dry around sections being careful not to touch the tissue and draw a circle around the sections with a Pap wax pen (*KYLIE: this reduces the amount of liquid reagents needed if you only have a few sections on the one slide, I have also put two different antibodies on the one slide (OR put antibody on one section only) BUT only if you are very careful and keep the solution in the circle.*)
- Add 50-100ul blocking buffer to each section, do not allow to dry out

- Incubate 1hr RT in a humidified chamber or O/N 4°C in a humidified chamber

Humidified chamber: we use large square flat tissue culture trays with two rows of 1ml plastic pipettes fixed to the bottom of the tray to keep the slides raised up, and place wet tissues in the bottom.

General:

KYLIE: I have been using 2% sheep serum (HIA) for blocking and 1%sheep serum or PBS for diluting the antibodies.

Alternatively, any of the following blocking buffers could be used:

- 1% - 10% sheep serum or goat serum in PBS
- 1% BSA in PBS
- Skim milk in PBS

M.O.M. kit:

M.O.M. kit: Mouse Ig Blocking Reagent: add 2 drops (90ul) of stock solution to 2.5ml PBS.

1. incubate 1hr in working solution of M.O.M. Mouse Ig Blocking Reagent.
2. Make M.O.M. Diluent:
add 600ul Protein Concentrate stock solution to 7.5ml PBS.
3. Wash 2 X 2min PBS
4. Incubate 5min in working solution of M.O.M. Diluent

Primary Antibody

The dilution of the primary antibody is dependent on the antibody, generally 1/100 in the blocking buffer used.

General:

- Tip off excess blocking buffer (do not wash)
- Add 50-100ul primary antibody diluted in blocking buffer
- Incubate 1hr RT or O/N 4C in a humidified chamber

M.O.M. kit:

1. Tip off excess M.O.M. Diluent
2. Dilute primary antibody in M.O.M. Diluent to the appropriate concentration

3. Incubate 30min
4. Make up VECTASTAIN Elite ABC Reagent:
Add 2 drops REAGENT A (grey label) to 5ml PBS in the ABC Reagent mixing bottle
Add 2 drops REAGENT B (grey) to the same mixing bottle, mix immediately and Allow to stand for about 30min before use.

Secondary Antibody

General:

- Tip off primary antibody (it is possible to re-use this solution if you need to)
- Wash 1 X 2 min PBS, 1 X 3min PBS
- Add 100µl secondary antibody (usually diluted 1/200 in PBS)
- Incubate 1hr RT or O/N 4°C in a humidified chamber
- Make up VECTASTAIN Elite ABC Reagent:
add 2 drops REAGENT A (grey label) to 5ml PBS in the ABC Reagent mixing bottle
add 2 drops REAGENT B (grey) to the same mixing bottle, mix immediately and allow to stand for about 30min before use.

MOM kit:

- wash 2 X 2min PBS
- Apply working solution of M.O.M. Biotinylated Anti-Mouse IgG Reagent
- i.e. 10ul stock solution to 2.5ml M.O.M. Diluent
- Incubate 10min

Colour Reaction

All applications including General and MOM kits:

- wash 1 X 2min PBS and 1 X 3min PBS
- incubate sections for 30min with VECTASTAIN Elite ABC Reagent
- make up peroxidase substrate solution

Vector:

To 5ml distilled water add 2 drops Buffer stock solution mix well.

Add 4 drops DAB stock solution mix well.

Add 2 drops hydrogen peroxide solution mix well.

Keep away from light and use within 30min ONLY (background is created if left any longer).

- wash 5min PBS
- incubate 2-10min in peroxidase substrate solution checking under a dissecting microscope the whole time
- rinse in PBS in coplin jar
- fix in 4% PFA in PBS 20min RT in fumehood
- rinse thoroughly in PBS with a few changes of solution

Mounting in aqueous mounting medium:

Tip off excess water and dry without touching the sections.

Add enough aqueous mounting medium to cover the slide and add coverslip.

Mounting in xylene-base mountant:

KYLIE: mounting in xylene mountant is best as it lasts a lot longer

5min 70% ethanol

5min 100% ethanol

10min xylene

Add enough xylene mounting medium to cover the slide and add coverslip.

Associated publications

Rumballe BA, Chiu HS, Georgas KM and Little MH.

Use of in situ hybridization to examine gene expression in the embryonic, neonatal and adult urogenital system.

Methods in Molecular Biology: Kidney Development (commissioned book chapter).

Rumballe B, Georgas K, Little MH.

[High-throughput paraffin section in situ hybridisation and dual](#)

immunohistochemistry

CSH Protocols, 2008

Georgas K, Rumballe B, Wilkinson L, Chiu HS, Lesieur E, Gilbert T, Little MH.

Use of dual section mRNA in situ hybridisation/immunohistochemistry to clarify gene expression patterns during the early stages of nephron development in the embryo and in the mature nephron of the adult mouse kidney.

Histochem Cell Biol. 2008 Nov;130(5):927-42. Epub 2008 Jul 11.